

Z-TRANSFORM

ENSC 429 – Digital Signal Processing

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Z-TRANSFORM DEFINITION

- z-transform is a generalization of DTFT
 - It exists even for some sequences for which DTFT does not exist
 - It is sometimes more convenient to work with than DTFT

- Recall the definition of DTFT:

$$\text{DTFT}\{x[n]\} = X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- z-transform is defined as:

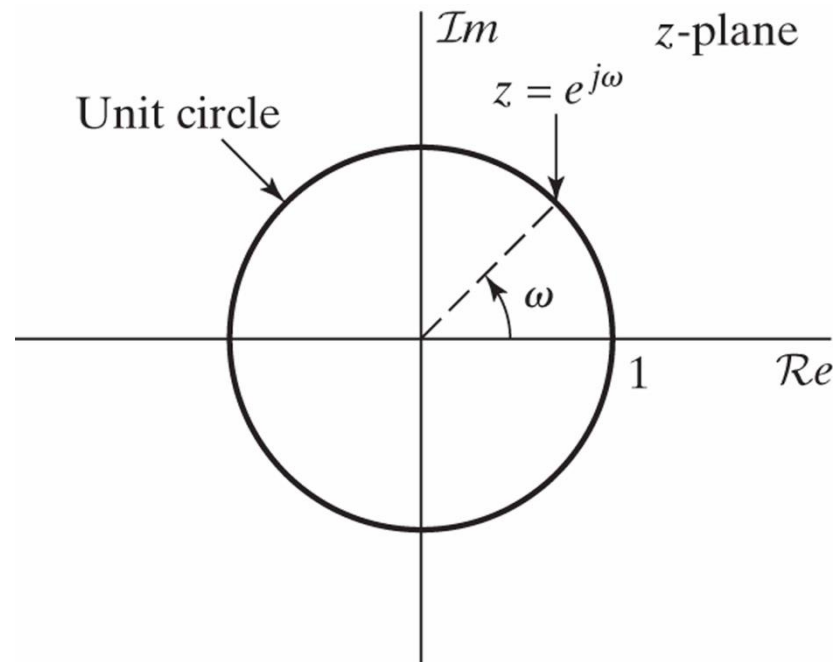
$$\mathcal{Z}\{x[n]\} = X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

where z is a complex number: $z = re^{j\omega}$

RELATIONSHIP TO DTFT

- Comparing the definitions of DTFT and z-transform, it is clear that DTFT is a special case of the z-transform when $r = 1$, that is, when $z = e^{j\omega}$

$$X(e^{j\omega}) = X(z) \Big|_{z=e^{j\omega}}$$

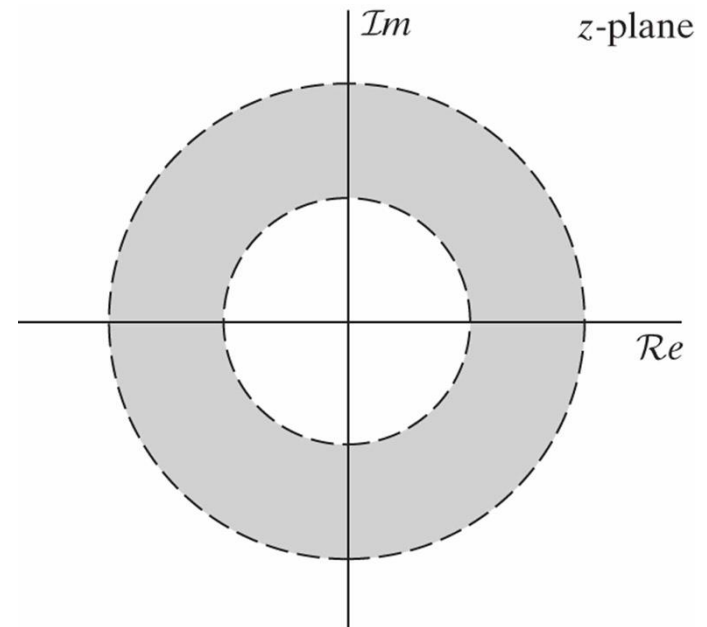


EXISTENCE OF THE Z-TRANSFORM

- z-transform exists when the following condition is met:

$$\begin{aligned} \left| \sum_{n=-\infty}^{\infty} x[n](re^{j\omega})^{-n} \right| &\leq \sum_{n=-\infty}^{\infty} |x[n]r^{-n}e^{-j\omega n}| = \sum_{n=-\infty}^{\infty} |x[n]r^{-n}| \\ &= \sum_{n=-\infty}^{\infty} |x[n]| \cdot |r|^{-n} < \infty \end{aligned}$$

- For a given $x[n]$, this depends on the value of $|r|$
- The set of values of $z = re^{-j\omega}$ for which the condition is met is called **region of convergence (ROC)**
- A typical ROC is a ring in the z-plane



EXAMPLE OF Z-TRANSFORM AND ITS ROC

Right-sided exponential sequence: $x[n] = a^n u[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

If it converges

Recall that a geometric series converges if $|r| < 1$:

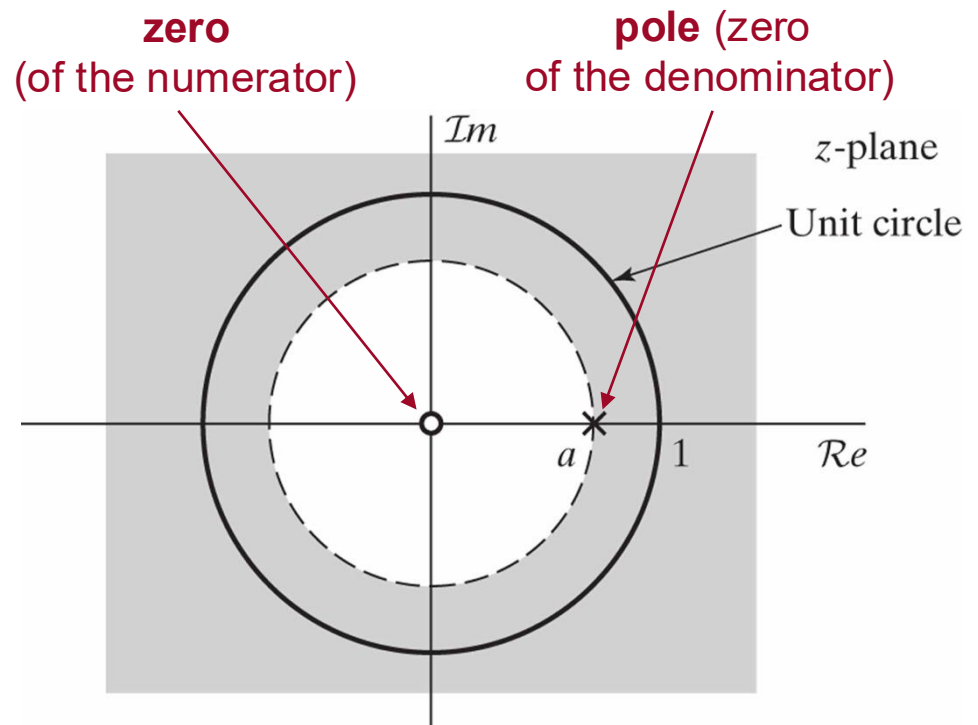
$$\sum_{n=0}^{\infty} r^n = \frac{1}{1 - r}$$

So the ROC for $X(z)$ above is

$$|az^{-1}| < 1$$

or, equivalently

$$|z| > |a|$$



EXAMPLE OF Z-TRANSFORM AND ITS ROC

Left-sided exponential sequence:

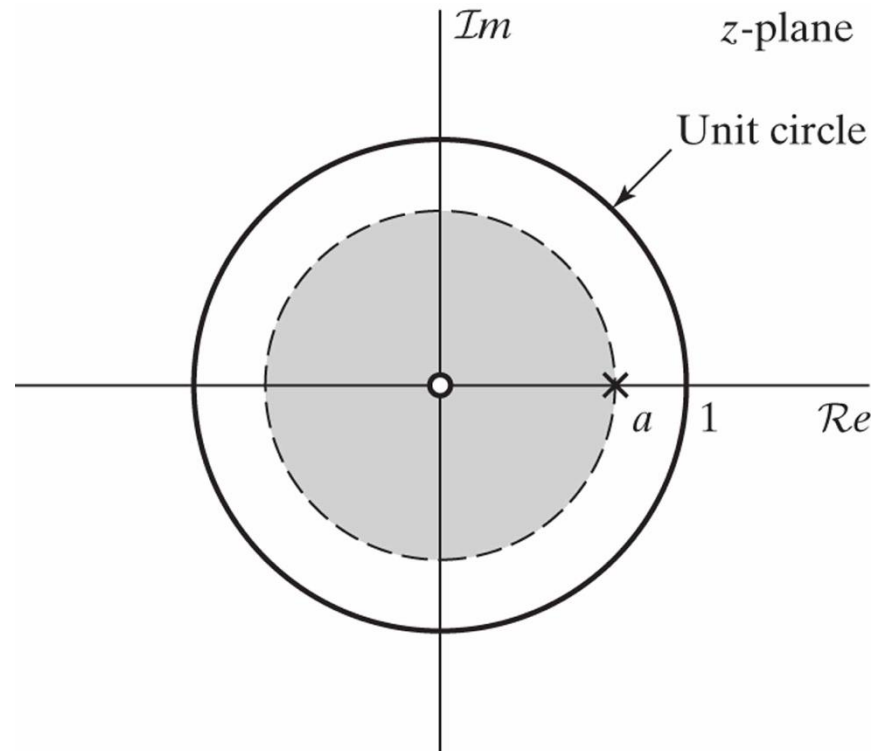
$$x[n] = -a^n u[-n - 1] = \begin{cases} -a^n & n \leq -1 \\ 0 & n > -1 \end{cases}$$

[See notes](#) for details

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

ROC is

$$|z| < |a|$$



EXAMPLE OF Z-TRANSFORM AND ITS ROC

Sum of two exponential sequences:

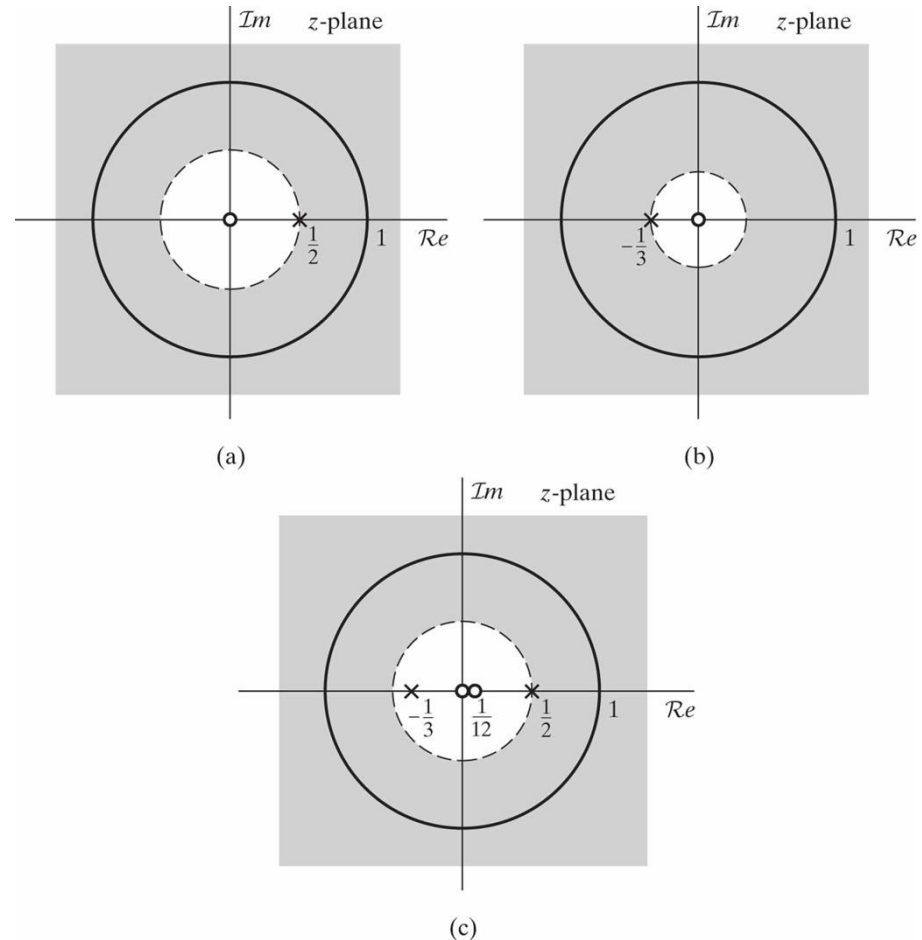
$$x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n]$$

See notes for details

$$X(z) = \frac{2z \left(z - \frac{1}{12}\right)}{\left(z - \frac{1}{2}\right) \left(z + \frac{1}{3}\right)}$$

ROC is

$$|z| > \frac{1}{2}$$



EXAMPLE OF Z-TRANSFORM AND ITS ROC

Two-sided exponential sequence:

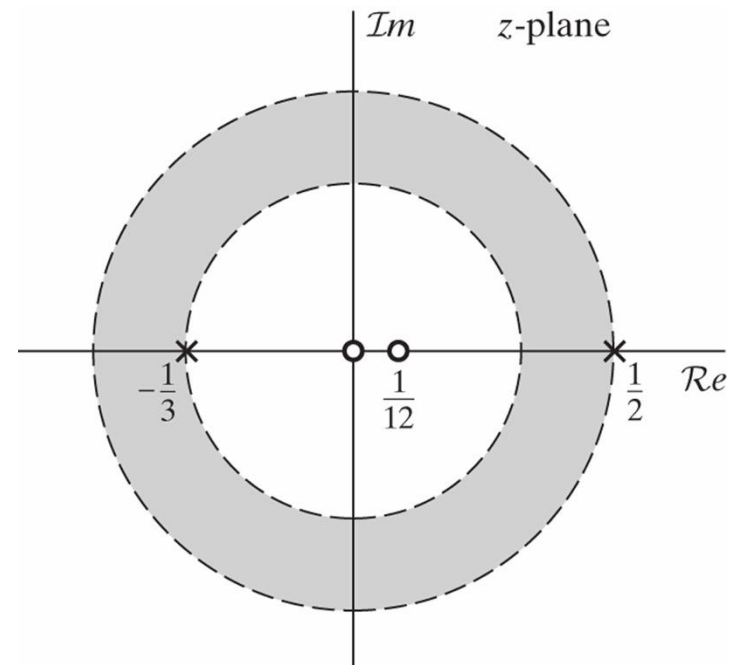
$$x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1]$$

See notes for details

$$X(z) = \frac{2z \left(z - \frac{1}{12}\right)}{\left(z - \frac{1}{2}\right) \left(z + \frac{1}{3}\right)}$$

ROC is

$$\frac{1}{3} < |z| < \frac{1}{2}$$



EXAMPLE OF Z-TRANSFORM AND ITS ROC

Truncated exponential sequence:

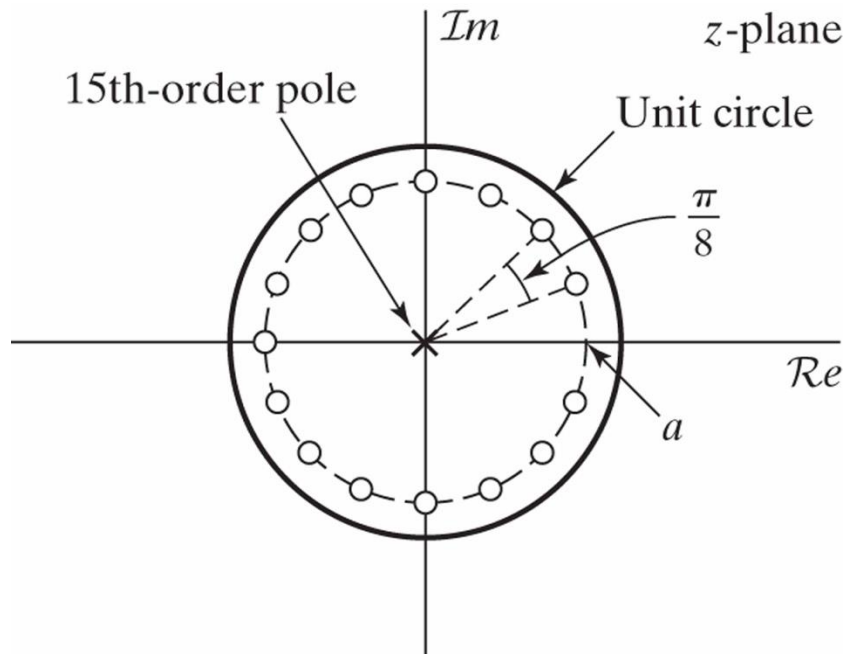
$$x[n] = \begin{cases} a^n & 0 \leq n \leq 15 \\ 0 & \text{else} \end{cases}$$

See notes for details

$$X(z) = \frac{1}{z^{15}} \frac{z^{16} - a^{16}}{z - a}$$

ROC is

$$0 < |z|$$



COMMON Z-TRANSFORM PAIRS

TABLE 3.1 SOME COMMON z -TRANSFORM PAIRS

Sequence	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3. $-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
4. $\delta[n - m]$	z^{-m}	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
6. $-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
7. $na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
8. $-na^n u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
9. $\cos(\omega_0 n)u[n]$	$\frac{1 - \cos(\omega_0)z^{-1}}{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}$	$ z > 1$
10. $\sin(\omega_0 n)u[n]$	$\frac{\sin(\omega_0)z^{-1}}{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}$	$ z > 1$
11. $r^n \cos(\omega_0 n)u[n]$	$\frac{1 - r\cos(\omega_0)z^{-1}}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}}$	$ z > r$
12. $r^n \sin(\omega_0 n)u[n]$	$\frac{r\sin(\omega_0)z^{-1}}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}}$	$ z > r$
13. $\begin{cases} a^n, & 0 \leq n \leq N - 1, \\ 0, & \text{otherwise} \end{cases}$	$\frac{1 - a^N z^{-N}}{1 - az^{-1}}$	$ z > 0$

PROPERTIES OF THE Z-TRANSFORM

- Assume z-transform is a rational function of the form

$$H(z) = \frac{P(z)}{Q(z)}$$

where $P(z)$ and $Q(z)$ are polynomials. Zeros of $P(z)$ are **zeros** of the transform, and zeros of $Q(z)$ are **poles**.

- This type of transform is the most common in DT system analysis and design
- It arises from the difference equation

$$a_0y[n] + a_1y[n-1] + \dots + a_Ny[n-N] = b_0x[n] + b_1x[n-1] + \dots + b_My[n-M]$$

- Applying the z-transform (remembering $\mathcal{Z}\{\delta[n-m]\} = z^{-m}$):

$$a_0Y(z) + a_1Y(z)z^{-1} + \dots + a_NY(z)z^{-N} = b_0X(z) + b_1X(z)z^{-1} + \dots + b_MX(z)z^{-M}$$

$$Y(z)[a_0 + a_1z^{-1} + \dots + a_Nz^{-N}] = X(z)[b_0 + b_1z^{-1} + \dots + b_Mz^{-M}]$$

$$\frac{Y(z)}{X(z)} = \frac{b_0 + b_1z^{-1} + \dots + b_Mz^{-M}}{a_0 + a_1z^{-1} + \dots + a_Nz^{-N}} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

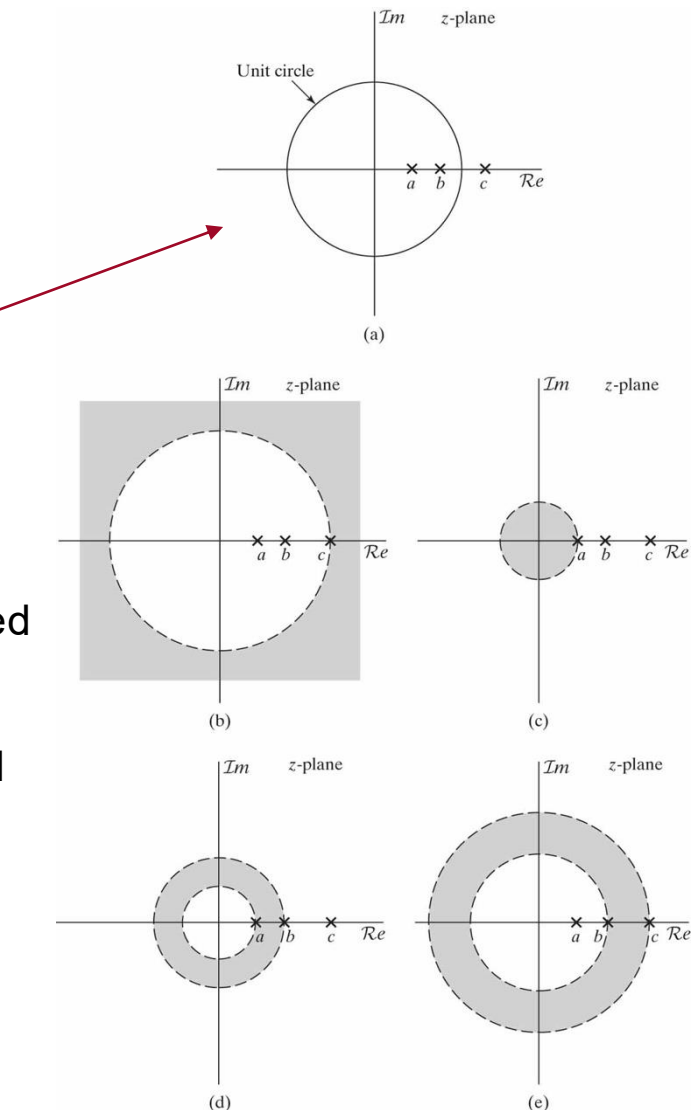
PROPERTIES OF THE Z-TRANSFORM

For a rational z-transform, the ROC has the following properties ([see notes](#) for illustrations)

Property	
1	ROC is of the form $0 \leq r_R < z < r_L \leq \infty$
2	DTFT exists if the ROC includes the unit circle
3	ROC cannot contain any poles
4	If $x[n]$ is a finite-duration sequence , its ROC is the entire z-plane, except possibly $z = 0$ or $z = \infty$
5	If $x[n]$ is a right-sided sequence , i.e., zero for $n < N_1 < \infty$, its ROC extends outward from the outermost pole to (and possibly including) $z = \infty$
6	If $x[n]$ is a left-sided sequence , i.e., zero for $n > N_2 > -\infty$, its ROC extends inward from the innermost pole to (and possibly including) $z = 0$
7	If $x[n]$ is a two-sided sequence , i.e., infinite duration, and neither left- nor right-sided, its ROC is a ring bounded by its poles
8	ROC is a connected region

PROPERTIES OF THE Z-TRANSFORM

- Based on the properties of ROC, for a given arrangement of poles, there are limited possibilities for a valid ROC
- For example, if there are three poles as shown, there are only four possibilities for a valid ROC
- DTFT will exist in only one of these four cases – the one where ROC includes the unit circle
- One of these cases corresponds to a right-handed sequence – which one?
- One of these cases corresponds to a left-handed sequence – which one?
- Two of the cases correspond to two-sided sequences – which ones?



INVERSE Z-TRANSFORM

- Recall: forward z-transform:

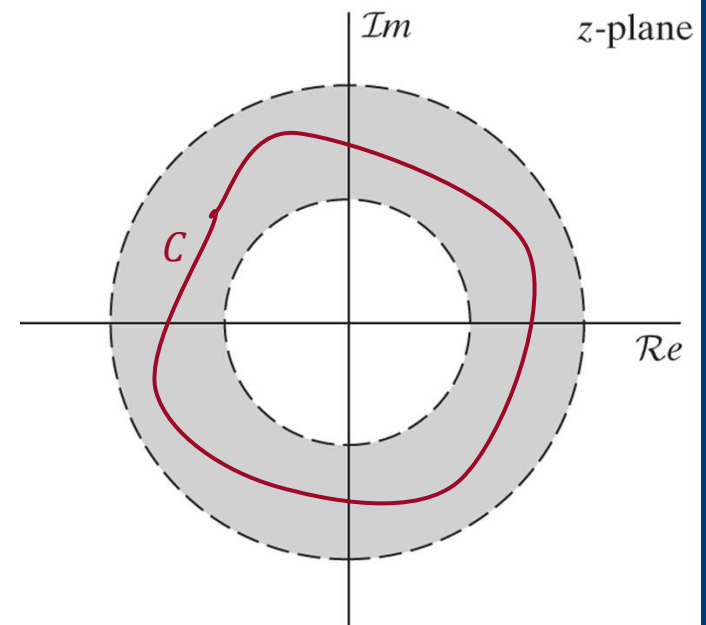
$$\mathcal{Z}\{x[n]\} = X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

- The inverse z-transform:

$$z[n] = \mathcal{Z}^{-1}\{X(z)\} = \frac{1}{2\pi j} \oint_C X(z)z^{n-1}dz$$

where C is a closed contour in the ROC in the complex z-plane

- This is an integral of a complex function in the complex plane, and it is studied in complex analysis
- The main result related to such integration is the Cauchy's Residue Theorem



INVERSE Z-TRANSFORM

- We will use less formal methods that work for the kinds of z-transforms we usually find in DT LTI system analysis
- These methods include:
 1. Inspection method
 2. Partial fractions expansion
 3. Power series expansion (long division)
- But always remember that these are special cases of the general inverse z-transform given on the previous slide

EXAMPLE OF INVERSE Z-TRANSFORM

- Suppose we need to find the inverse z-transform of

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

- By inspecting Table 3.1 (slide 10), we find the following z-transform pair

$$a^n u[n] \xleftrightarrow{z} \frac{1}{1 - az^{-1}}, \quad |z| > a$$

- We recognize that this matches the above z-transform with $a = \frac{1}{2}$, so

$$x[n] = \mathcal{Z}^{-1}\{X(z)\} = \left(\frac{1}{2}\right)^n u[n]$$

- Had the ROC been $|z| < \frac{1}{2}$, this would have matched a different pair in the table, and the inverse z-transform would have been $x[n] = -\left(\frac{1}{2}\right)^n u[-n - 1]$

INVERSE Z-TRANSFORM BY PARTIAL FRACTION EXPANSION

- Consider the rational z-transform:

$$X(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

- Partial fraction expansion is about expressing the above rational function as the sum of simpler fractions that exist in Table 3.1
- Note that both numerator and denominator of $X(z)$ are polynomials in z^{-1} , so they can be factorized as

$$X(z) = \frac{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}{a_0 \prod_{k=1}^N (1 - d_k z^{-1})}$$

where c_k and d_k are roots of the numerator and denominator (i.e., zeros and poles of $X(z)$), respectively

INVERSE Z-TRANSFORM BY PARTIAL FRACTION EXPANSION

$$X(z) = \frac{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}{a_0 \prod_{k=1}^N (1 - d_k z^{-1})}$$

- The most general form of expansion of this expression that we will encounter is

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k: d_k \text{ first-order pole}} \frac{A_k}{1 - d_k z^{-1}} + \sum_{k: d_k \text{ pole of order } s_k > 1} \sum_{m=1}^{s_k} \frac{C_{k,m}}{(1 - d_k z^{-1})^m}$$

Exists only when $M \geq N$.
In such cases, B_r 's can
be obtained by long
division.

Exists when there
are higher-order
poles.

INVERSE Z-TRANSFORM BY PARTIAL FRACTION EXPANSION

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{\substack{k: \\ d_k \text{ first-order pole}}} \frac{A_k}{1 - d_k z^{-1}} + \sum_{\substack{k: \\ d_k \text{ pole of order } s_k > 1}} \sum_{m=1}^{s_k} \frac{C_{k,m}}{(1 - d_k z^{-1})^m}$$

- B_r obtained by long division

$$A_k = (1 - d_k z^{-1}) X(z) \Big|_{z=d_k}$$

$$C_{k,m} = \frac{1}{(s_k - m)! (-d_k)^{s_k - m}} \left\{ \frac{d^{s_k - m}}{dw^{s_k - m}} [(1 - d_k w)^{s_k} X(w^{-1})] \right\}_{w=d_k^{-1}}$$

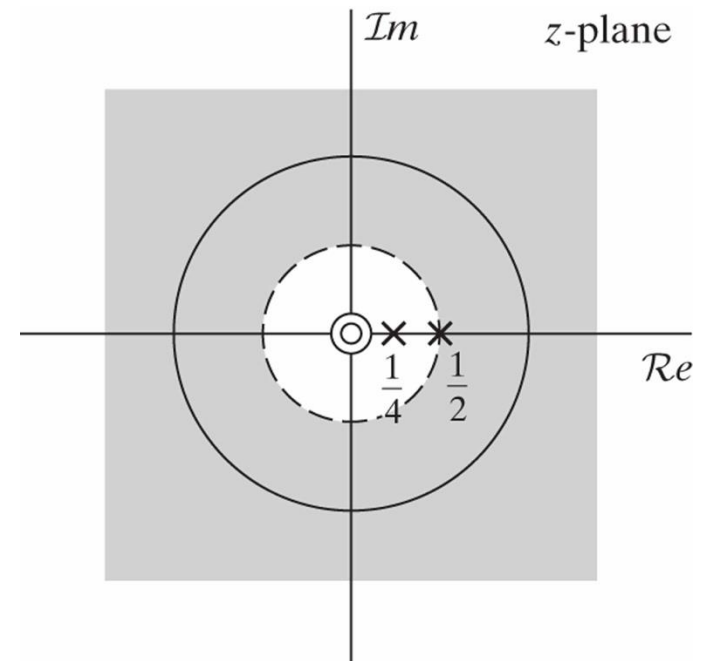
EXAMPLE OF INVERSE Z-TRANSFORM

Find the inverse z-transform of

$$X(z) = \frac{1}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{2}z^{-1}\right)}, \quad |z| > \frac{1}{2}$$

(See notes for details)

$$x[n] = 2\left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$



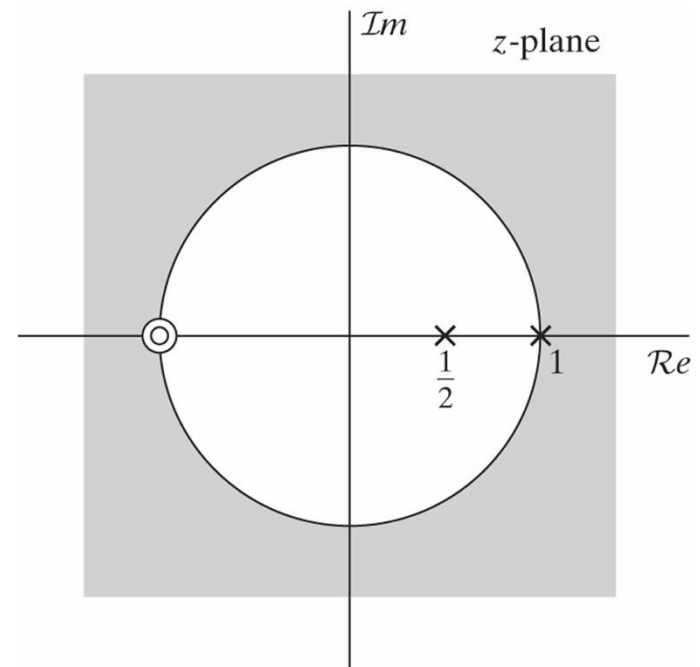
EXAMPLE OF INVERSE Z-TRANSFORM

Find the inverse z-transform of

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}, \quad |z| > 1$$

([See notes](#) for details)

$$x[n] = 2\delta[n] - 9\left(\frac{1}{2}\right)^n u[n] + 8u[n]$$



INVERSE Z-TRANSFORM BY POWER SERIES EXPANSION

- Recall the definition of z-transform:

$$\mathcal{Z}\{x[n]\} = X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

- If the z-transform is given in this form, we can read off the values of $x[n]$ directly
- Formally, since $\delta[n - m] \xleftrightarrow{Z} z^{-m}$, we have

$$x[n] = \sum_{m=-\infty}^{\infty} x[m]\delta[n - m]$$

- Example:

$$X(z) = z^2 - \frac{1}{2}z - 1 + \frac{1}{2}z^{-1}$$
$$x[n] = \delta[n + 2] - \frac{1}{2}\delta[n + 1] - \delta[n] + \frac{1}{2}\delta[n - 1]$$

INVERSE Z-TRANSFORM BY POWER SERIES EXPANSION

- Using power series, we can also handle z-transforms that are not rational functions
- Example:

$$X(z) = \ln(1 + az^{-1}), \quad |z| > |a|$$

- Recall the power series expansion of $\ln(1 + x)$ for $|x| < 1$:

$$\ln(1 + x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$$

- Hence, with $x \equiv az^{-1}$, we have

$$X(z) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{a^n z^{-n}}{n} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{a^n}{n} z^{-n}$$

- So

$$x[n] = (-1)^{n+1} \frac{a^n}{n} u[n-1]$$

INVERSE Z-TRANSFORM BY POWER SERIES EXPANSION

- Another example:

$$X(z) = \frac{1}{1 + az^{-1}}, \quad |z| > |a|$$

- This expression is in Table 3.1, but doing partial fraction expansion by long division shows where the solution came from
- [See notes](#) for details

$$x[n] = a^n u[n]$$

Z-TRANSFORM PROPERTIES

TABLE 3.2 SOME z -TRANSFORM PROPERTIES

Property Number	Section Reference	Sequence	Transform	ROC
		$x[n]$	$X(z)$	R_x
		$x_1[n]$	$X_1(z)$	R_{x_1}
		$x_2[n]$	$X_2(z)$	R_{x_2}
1	3.4.1	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	Contains $R_{x_1} \cap R_{x_2}$
2	3.4.2	$x[n - n_0]$	$z^{-n_0}X(z)$	R_x , except for the possible addition or deletion of the origin or ∞
3	3.4.3	$z_0^n x[n]$	$X(z/z_0)$	$ z_0 R_x$
4	3.4.4	$nx[n]$	$-z \frac{dX(z)}{dz}$	R_x
5	3.4.5	$x^*[n]$	$X^*(z^*)$	R_x
6		$\mathcal{Re}\{x[n]\}$	$\frac{1}{2}[X(z) + X^*(z^*)]$	Contains R_x
7		$\mathcal{Im}\{x[n]\}$	$\frac{1}{2j}[X(z) - X^*(z^*)]$	Contains R_x
8	3.4.6	$x^*[-n]$	$X^*(1/z^*)$	$1/R_x$
9	3.4.7	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	Contains $R_{x_1} \cap R_{x_2}$

Z-TRANSFORM PROPERTIES

- Most properties can be proved from the definition of the z-transform
- Let us prove the convolution property:

$$x_1[n] * x_2[n] \xrightarrow{Z} X_1(z)X_2(z), \quad \text{ROC} \supseteq R_{x_1} \cap R_{x_2}$$

- [See notes](#) for details

Z-TRANSFORM AND LTI SYSTEMS

- Due to the convolution property of the z-transform, we can analyze input-output relationships of an LTI system in the z-transform domain

$$y[n] = x[n] * h[n] \implies Y(z) = X(z)H(z)$$

- Sometimes, taking the z-transform, doing the multiplication in the z-domain, and taking the inverse z-transform, might be simpler than convolution in time domain
- Example:

$$x[n] = Au[n], \quad h[n] = a^n u[n], \quad |a| < 1$$

- [See notes](#) for details

$$y[n] = x[n] * h[n] = \frac{A}{1-a} (1 - a^{n+1})u[n]$$

READING ASSIGNMENT

- Read Sections 3.1 – 3.5 in the textbook
- Start reading Chapter 4